

Marv Predict Max - All Backtest Results

2026-10-09

Summary table (walk-forward, seasons the models did not train on)

Sport	Test seasons	All games	Experts agree	80%+ confidence	85%+ confidence
NCAA men's	2023-25	73.9%	78.0%	90.8% (~2,300/yr)	93.9% (~1,700/yr)
NCAA women's	2023-25	77.4%	81.9%	91.5% (~2,700/yr)	94.2% (~2,150/yr)
WNBA	2023-25	66.6%	70.0%	84.1% (~38/yr)	87.2% (~16/yr)
NBA	2024-26	66.8%	70.0%	84.0% (~225/yr)	87.2% (~104/yr)
NFL	2023-25	64.8%	68.1%	79.5% (~26/yr)	95.5% (22 picks)
College football	2023-25	77.0%	78.8%	88.3% (~134/yr)	91.9% (~102/yr)

College basketball rows include lopsided games against lower-division teams; the D1-vs-D1-only figures were about 90.3% (men's) and 90.8% (women's). These are mostly heavy favorites, so profit after the vig is about break-even where lines exist.

Market vs model (NFL 2018-25, n=2,219; college football 2018-25, n=2,336)

Test	Result
Model beyond the closing spread (stacked logistic)	No. NFL model coefficient 0.00-0.01 vs market 0.14; college football 0.00 vs 0.12
Brier score NFL	market 0.2108, stack 0.2109, model alone 0.2204
Totals: model beyond the line	No. Top-20% edge hit 50.2% (NFL), 50.4% (college football)
NFL top 400 moneyline picks	model alone 76.5% (ROI -6.6%); market 84.3% (-0.4%); 20% model + 80% market 84.8% (+0.2%)
85%+ confidence	NFL 88.6-90.5%, college football 93.4% (900 games) vs 88.6% / 91.5% model-only

Engine change: stats-mode moneyline confidence for NFL and college football is now 0.2 x model + 0.8 x no-vig market, floor 85%.

Pending VM backtests

College men's (2019-2025): no result, job killed after about 80 minutes (likely memory). College women's: running at time of writing. WNBA: held. These add historical-line ROI and over/under results. Nothing here is a proven profit edge; all signals are paper-tracked.

Full analysis write-up (ANALYSIS.md)

Analysis of the Marv Predict Max documents

Five PDFs were reviewed: College Football Predict Max, the Predict Bot System Blueprint (CFB metric ledger), Soccer Predict Max (Expanded), Hockey Predict Max and Pro Baseball Predict Max.

The headline problem: the backtest numbers are not achievable

Document	Claimed result	Reality check
CFB Predict Max	100% spread, 100% O/U on 27/24 plays (3 weeks)	Sharp, profitable bettors hit ~54-58% ATS long-term. Books shade lines to ~50/50.
CFB "old linear" baseline	86.7% ATS, 80% O/U on 30 games	Also far above any known public or professional model.
Baseball Predict Max	97-100% moneyline, 96-100% O/U, 188 plays	MLB favorites rarely exceed 70% true win probability; a 98% ML hit rate is not possible without information leakage.

A 100% (or 98%) hit rate on a few dozen bets is the signature of **look-ahead leakage or in-sample fitting**, not of a real edge. Common causes:

1. Ratings built from season-long stats that include the games being "predicted".
2. Thresholds or vetoes chosen after seeing which games won (the veto "passes" the losers).
3. Grading against closing lines while the model saw post-game or closing information.
4. Tiny samples: 24-30 plays over 3 weeks can't distinguish 60% skill from luck, let alone prove 100%.

The Soccer and Hockey specs get this right ("Never force a 90% hit rate; report the genuine out-of-sample result", strict point-in-time data, CLV/Brier/calibration tracking). **That standard should apply to all of the Marv models**, including the CFB and baseball ones. Realistic targets: 53-57% against the spread or totals, positive closing-line value (CLV) and calibrated probabilities. Anything above about 60% sustained should be treated as a bug to hunt down.

Document-by-document

CFB Predict Max: a sound core idea (Monte Carlo instead of linear point estimates, explicit garbage-time and pace handling, and a veto layer that passes 20-30% of the slate). It is under-specified: no data sources, rating method, simulation parameters, edge thresholds or definition of a "trap line". The bot in this repo fills those gaps with concrete, testable choices (see README).

System Blueprint (38-metric ledger): counting "categories won" throws away magnitude (winning rush defense by 0.1 yd counts the same as by 80 yds), double-counts correlated stats (PPG, total offense and PPA all measure the same thing) and the Top-30 opponent filter leaves 2-4 games per team, which is too few. It's better used as a feature list feeding opponent-adjusted ratings than as a scoring system. Note that standard CFB home-field advantage has fallen to ~2-2.5 points, not 3.

Soccer Predict Max (Expanded): the most rigorous spec, with a frozen architecture, Dixon-Coles scoreline matrix, markets derived from one distribution and walk-forward validation. The blocker is data: xG, PPDA, lineups and keeper PSxG need a paid feed (e.g. Opta/Stats Perform, Sportmonks, API-Football) or scraping sites whose terms forbid it.

Hockey Predict Max: also rigorous. The goalie-confirmation gate, empty-net modelling and separate regulation and OT outcomes are the right priorities. The free NHL API (api-web.nhle.com) covers schedules, scores, standings and probable goalies. Odds would come from The Odds API.

Baseball Predict Max: the plate-appearance Monte Carlo plus a 60-65% veto rate is a reasonable design, but the reported 97-100% results need to be discarded and re-tested walk-forward. MLB Stats API (free) and Statcast/pybaseball cover the data; odds from The Odds API.

Honest NFL backtest of this bot

nflverse has every NFL game since 1999 with closing lines, so the bot was tested walk-forward: each week was predicted using only earlier games and graded against the real closing line.

Seasons	Spread (after veto)	Totals (after veto)	Moneyline (after veto)	Brier: model vs market
2018-2023 (used for tuning)	128-130, -2.9% ROI	145-173, -10.6% ROI	19-34, -19.4% ROI	—
2024-2025 (held out)	41-37, +1.3% ROI	47-42, +1.5% ROI	10-12, +3.0% ROI	0.2205 vs 0.2059

What this means:

- **The market beats a scores-only model.** Closing-line win probabilities are more accurate than the model’s (lower Brier). Across 2018-2023, weighting the model more never improved on the market alone.
- **The veto stack works as a filter**, not as an edge generator. It passed 85-89% of the slate and cut losses versus betting every model edge (spread ROI -4.1% → -2.9%; moneyline bets cut from 1,657 to 53), but six seasons of results don’t show a profitable edge.
- The small 2024-2025 profit is within noise for ~80 bets per market.
- Longshot moneylines were the biggest leak (the favorite-longshot bias), so NFL underdogs are capped at +150 and other sports at +250.

That’s why the bot ships in **paper-trading mode**: it alerts and grades every pick, and a real edge has to show up in its own ledger (ideally including beating the closing line) before money goes on it. Edges more realistically come from information the market hasn’t priced yet: early lines before they move, confirmed lineups, goalies and pitchers, injuries and weather. The specs list these, but they need paid data feeds.

The other sports couldn’t be backtested from the build environment (ESPN wasn’t reachable there). Run `python -m marv backtest` for each sport on the VM.

Stats-expert backtests (every offense & defense stat, random forest, consensus)

NFL, NBA, WNBA and MLB now run on full box scores: every offensive, defensive and special-teams column the free data has (131 NFL team stats incl. EPA and passer rating; NBA/WNBA box scores plus eFG%, TOV%, FT rate, rebound % and offensive/defensive rating per 100 possessions; MLB batting, pitching and fielding lines plus starting-pitcher form and park factors). Three experts (a ridge “stat formula”, a random forest and opponent-adjusted power ratings) each project the score; the consensus feeds the Monte Carlo.

Everything is walk-forward (features use only earlier games; models are refit monthly on the past). Thresholds were picked on the early seasons and then measured on later **test** seasons the tuning never saw:

Sport (test seasons)	Every game: winner	Moneyline floor	Test picks	Test accuracy	Test ROI	Over/under (test)
NBA (2022-26)	65.4% (favorite 67.8%)	86%	126	90.5%	-1.2%	52.8% at 17% edge (ROI +0.9%)
NFL (2022-25)	65.8% (favorite 67.6%)	78%	30	90.0%	+2.9%	~49-51%, no edge

Sport (test seasons)	Every game: winner	Moneyline floor	Test picks	Test accuracy	Test ROI	Over/under (test)
WNBA (2022-25)	66.0%	85%	26	~83-88%	n/a (no odds data)	no totals data
MLB (2023-25)	56.4%	70%	43	76.7%	n/a (no odds data)	no totals data
NCAA men's basketball (2023-25, D1 vs D1)	71.4%	79%	4,260	90.3%	n/a here (VM pulls lines)	needs historical lines (VM)
NCAA women's basketball (2023-25, D1 vs D1)	76.2%	76%	7,076	90.8%	n/a here (VM pulls lines)	needs historical lines (VM)
College (top 30)	needs a CFBD key: run python -m marv stats-backtest --sport cfb --seasons 2016-2025 on the VM					

What this means:

- **90% moneyline accuracy is achievable in the NBA and NFL**, and held up on unseen seasons. It comes from backing heavy favorites only when all three experts agree and the model is very confident: about 25 NBA and 6 NFL picks a season. These are typically -400 to -1000 favorites, so the 90% hit rate is roughly break-even after the vig; it is not a money machine.
- **Over/under can't be pushed to 90%**. NFL totals sit at a coin flip at every edge; NBA totals show a small edge (54% on tuning seasons, 52.8% on test). Totals are only played for the NBA.
- **MLB tops out around 75%** even on its most confident picks. Baseball is the least predictable major sport (even betting favorites win only ~58-60%). Claims of 90%+ MLB moneyline or totals accuracy, like the 2025 postseason figures in the PDFs, are not achievable without the backtest seeing the results.
- **College basketball** has such large talent gaps that 90% moneylines are common (about 1,400 men's and 2,400 women's picks a season), all on strong favorites. Strength-of-schedule adjusted stats made the stat formula clearly better than power ratings alone (men's 73.8% vs 71.0% winners; women's 76.8% vs 73.2%). Profit after the vig still has to be checked with historical lines on the VM.
- The stat formula (every stat weighted) was the single best expert in every sport; the random forest alone was the weakest, but adding it to the consensus helped calibration slightly.

What was built

One multi-sport engine (marv/) covering NFL, college football, NBA, WNBA, NHL, MLB and soccer, with sport-specific simulators, the shared veto stack, Telegram cards, a graded pick ledger and a daily systemd timer for a Google Compute Engine VM. See README.md.

WNBA “Hybrid H2H Matrix & Trend Catcher” report (2026 playoffs) vs Marv

The report replaces model side-picks with a winner-take-all tally across every offense, defense, shooting and ball-security category, adjusted by a 3-game “trend catcher” (turnover spikes and rebounding drops versus the season

average). It claims ~78% ML/ATS and ~65% O/U, but it gives no sample, weights or seasons. Its “backtest” is three 2026 first-round series described after the fact.

Walk-forward test on every WNBA playoff game 2012-2026 (271 games). All features use only games played before tip-off, and the trend weight was fit on 2012-19 only:

Playoff winner accuracy	Home team	Marv (current)	H2H tally	H2H + trend (report)
2012-2026 (271)	62.0%	65.3%	61.3%	61.6%
2020-2026 held out (133)	63.9%	68.4%	60.9%	60.9%
2025 (24)	62.5%	58.3%	37.5%	37.5%
2026 so far (12)	83.3%	75.0%	50.0%	41.7%

- The category tally does worse than simply picking the home team. Counting categories throws away how big each edge is and counts correlated stats (FG%, eFG%, points, offensive rating) several times.
- The trend catcher gave no measurable lift. Larger trend weights made it worse (55-59%), and adding it to Marv’s margin chose a weight of zero. Three-game swings in WNBA box scores are mostly noise.
- Totals: the possession-loop total (13.6 MAE) was no better than Marv’s (13.3 MAE). 50.6% of playoff games finished under Marv’s number, so there’s no O/U edge in either direction (historical lines needed for a true O/U win rate; run on the VM). The report’s 195.5-197.5 Liberty-Dream total is far outside the WNBA range: both models projected ~175, and the game finished 174.
- Marv’s 85% WNBA moneyline floor produced zero playoff picks (playoff games are close). Its 70%+ playoff picks went 85.1% (40/47), the closest honest analogue to the report’s claim.

Verdict: keep Marv’s engine. Don’t adopt the tally or trend modifier.

Max Pick head-to-head method (stat-by-stat tally + last-3 trend + Monte Carlo)

python -m marv h2h-backtest --sport nfl|cfl|... --seasons 2016-2025. Every offense stat and every stat allowed on defense is compared team vs team: a point to the better team (e.g. yards per play allowed 4.4 vs 5.0). Which way is “better” is learned from earlier seasons. The same tally is repeated on each team’s last 3 games, last season’s averages carry in for early weeks, and the tally becomes a margin that the shared Monte Carlo turns into win, cover and over/under probabilities. College football now gets its stats free from play-by-play (yards per play, EPA, success rate, explosiveness, third downs, red zone, field position, turnovers...), so it no longer needs a CFBD key.

Walk-forward, same games, Marv vs the head-to-head method (held-out seasons):

	Marv ML (all / 80%+ picks)	H2H ML (all / 80%+ picks)	Spread best	O/U best
NFL 2022-25	66.2% / 100% (15)	63.7% / 75.8% (33)	~53-55%, noisy	~50%
College FB top 30, 2021-25	75.6% / 90.2% (285)	73.2% / 84.9% (351)	~49-52%	~50-51%
WNBA 2022-26	68.0% / 85.2% (108)	66.5% / 85.2% (142)	no lines	no lines
NBA 2022-26	65.4% / 84.2% (431)	64.0% / 83.2% (321)	n/a	~50%

The tally is close to Marv but slightly worse everywhere: counting categories throws away how big each edge is. Neither method beats closing totals.

2026, last 3 finished weeks (rules fixed on 2017-2025 first, nothing re-tuned on these games):

	Marv ML 70%+	H2H ML 70%+	Marv strict rule	ML every game	O/U
NFL weeks 2-4 (48 games)	87.5% (7/8)	81.8% (9/11)	1/1	58.3%	39.6%
College FB top 30, weeks 4-6 (64 games)	91.9% (34/37)	89.5% (34/38)	93.3% (14/15)	81.2%	51.6%

The college football picks hit the 90% target on unseen games. NFL is close on very few picks. Tuning until these specific games reach 90% would only fit the past and wouldn't carry forward.

NFL player props

marv/props/: passing, rushing and receiving yards plus receptions. Projection = player form (weighted recent, season, last 3), usage (attempts, carries, targets, target and air-yards share), the opponent's yards allowed to that position vs the league, and the implied team total from the spread and total. A gradient-boosted model per market, then a Monte Carlo drawn from the model's own past misses gives P(over). Picks need edge $\geq 5\%$ over the price's implied probability (probabilities pulled 30% toward 50% until the real-line backtest calibrates them).

Walk-forward 2022-2026 (free data, no real lines available here):

Market	Avg miss: model / season avg / recent form	vs an average-based line, edge $\geq 10\%$	always-under on that line
Passing yards	65.0 / 65.8 / 66.0	63.0% (1,293)	50.8%
Rushing yards	27.4 / 28.0 / 27.2	60.8% (1,709)	58.1%
Receiving yards	25.7 / 27.4 / 26.3	65.0% (3,121)	58.7%
Receptions	1.8 / 1.9 / 1.8	65.3% (3,244)	57.6%

The model clearly beats the average-based approach of props_analyzer.py (which went 47-51%), but these lines are naive: averages sit above typical outputs, so "always under" already wins ~58% outside passing yards. Sportsbook lines are set near the median and are much sharper, so expect far lower real win rates. The real test is props-backtest --real on the VM (Odds API historical props from 2023).

NCAA men's basketball player props

marv/props/ncaab.py: points, rebounds, assists and made threes for Division I players (20+ minutes). Projection = player form (weighted recent, season, last 3), minutes, shot volume and starter rate, the opponent's output allowed to that position group vs the league, both teams' pace and scoring, home/away. Same gradient-boosted model and Monte Carlo of past misses as the NFL props.

Walk-forward 2022-2026 (285,559 player-games per market; no real lines available here):

Market	Avg miss: model / season avg / recent form	vs average-based line, edge $\geq 10\%$	always-under	props_analyzer rule
Points	4.63 / 4.73 / 4.72	62.6%	52.3%	50.2%
Rebounds	1.90 / 1.95 / 1.93	63.6%	53.6%	52.8%
Assists	1.21 / 1.24 / 1.23	65.2%	54.8%	55.4%
Made threes	0.85 / 0.87 / 0.86	72.7%	62.6%	62.2%

2025-26 season alone, points at a 10% edge: 62.8%. Points is the most believable signal (only ~52% from leaning under). Real sportsbook lines are sharper than these average-based lines; the real test is props-backtest -- sport ncaab -- real on the VM, limited to games involving a top-50 team (a full D1 season of props would cost ~200k credits).

Quarter-by-quarter in-game model (walk-forward, held-out seasons)

At the end of each quarter (half in NCAA men's), the model projects the final margin and total from the pregame line, the score, and the game's own stats so far. It's a ridge correction on top of the original score-and-pace method; a model trained from scratch on game stats (boosted trees) did worse than the original method and was dropped.

Sport (games per checkpoint)	Checkpoint	Winner: stats model / original	Brier: stats / original	Final-total miss (pts): stats / original
NFL 2020-26 (1,752)	Q1 / Half / Q3	69.3 / 69.2 · 77.1 / 77.3 · 82.2 / 82.0%	0.196 / 0.196 · 0.156 / 0.157 · 0.126 / 0.126	9.6 / 10.5 · 7.8 / 7.9 · 6.2 / 6.4
College FB 2020-26 (3,774)	Q1 / Half / Q3	77.3 / 77.4 · 81.5 / 81.4 · 86.3 / 86.4%	0.153 / 0.153 · 0.127 / 0.126 · 0.096 / 0.097	11.1 / 12.0 · 8.8 / 9.0 · 6.9 / 6.9
WNBA 2021-26 (1,626)	Q1 / Half / Q3	71.2 / 70.0 · 76.6 / 77.3 · 82.8 / 82.5%	0.187 / 0.192 · 0.153 / 0.154 · 0.117 / 0.118	11.9 / 12.3 · 9.9 / 10.0 · 7.3 / 7.3
NCAA men's 2022-26 (29,944)	Half	80.3 / 78.8%	0.135 / 0.142	9.4 / 10.3
NCAA women's 2022-26 (27,795)	Q1 / Half / Q3	78.8 / 77.9 · 82.8 / 82.5 · 87.6 / 87.3%	0.143 / 0.149 · 0.117 / 0.120 · 0.086 / 0.087	11.2 / 11.6 · 9.0 / 9.5 · 6.7 / 6.8

- **College basketball is where game stats help:** shooting, rebounding, turnovers and pace at halftime add 1.5 points of winner accuracy and nearly a point off the total projection on ~30,000 games.
- **Football gains come from recalibration, not stats:** a score-only version of the same correction does as well, so yards per play and turnovers add nothing beyond the score.
- The over/under rates in the backtest output are against the *pregame* total, which is easy once the score is known; real live totals move with the score, and no historical live lines were available to test against.

Stress tests (all findings)

Player props, against a sportsbook-style line. The “book” here is a competent baseline projection (player form and usage, regressed) hung at its median, which is how books set lines. As a control, a model trained on shuffled outcomes scores about 50% against this line. Held-out seasons, edge >= 5%:

Sport	Market	Marv win rate (bets)	95% interval	Every season
NFL 2024-26	pass yds / rush yds / rec yds / receptions	59.2% / 57.1% / 57.1% / 59.5%	55-62%	55-62%
College FB 2023-26	pass / rush / rec / receptions	57.9% / 57.2% / 55.1% / 57.3%	53-60%	53-59%
WNBA 2023-26	points / rebounds / assists	58.8% / 58.8% / 59.1%	57-60%	57-61%
NCAA women's 2024-26	points / rebounds / assists	58.8% / 59.0% / 62.8% (70k-108k bets)	58-63%	flat
NCAA men's 2024-26	points / rebounds / assists	58.0% / 59.4% / 62.2% (68k-116k bets)	58-62%	flat

- **90% is never reached on a real-looking line.** The best validation slices topped out at 70-85% on a few hundred bets and gave back several points on held-out seasons.
- **Made threes looked like 97-98%. That's an artifact, not an edge:** they are unders at 0.5 for players who rarely shoot threes. A book won't hang that line at -110. Live props now cap probabilities at 15-85%.
- **What the numbers mean.** Marv's edge over a *simple* book is real and stable. Bovado prices with more information (injuries, minutes news, sharp action), so the real win rate will be lower. Only props -backtest - real on the VM measures that. Even 54-55% at -110 would be profitable.

Game moneyline picks (the ~90% findings). Bootstrap intervals, season-by-season results and threshold sensitivity hold up for college football (91.7%, 95% interval 88-96%) and NCAA men's (90.2%, 89-91%). NFL (28 picks) and WNBA (61) are too small to be sure: lowering the NFL cutoff from 78% to 74% drops it to 82%. The market's own biggest favorites, taken in the same numbers, did as well or better (college FB 94.5%, NFL 100%). These picks are heavy favorites the market already prices in.

Recommendation engine on real closing prices. No game market (NFL moneyline/spread/total, college football spread/total, NBA moneyline/spread/total) produced a positive, statistically significant held-out ROI. NBA totals came closest (+9.4% on 157 bets, $p = 0.11$). So the board shows game markets as leans at most and hides the ones that lost money.

Bugs the stress tests caught and fixed: NBA spreads before 2022-23 were unsigned in the source file (about a third of games had the favorite backwards), and average-based "lines" were beatable even by a model that knew nothing, which inflated earlier props results.

Elimination filters, injuries/roster, weather and home advantage

"Eliminate a pick if the favorite doesn't meet expectations" (the Gemini idea) and every other filter was chosen on early seasons and measured on later ones the filters never saw.

NFL (picks chosen on 2016-21, tested on 2022-26; ROI at real closing moneylines):

Rule	Held-out accuracy	Picks/season	ROI
Marv 70%+	75.4%	35	-8.4%
+ drop if 1.5+ full-time starters out, wind 15+ mph, or market under 70%	78.4%	23	-7.0%
Marv agrees with the market and market 85%+	93.4%	15	+3.5% (avg price -950)
best rules that hit 90% on 2016-21	85-87%	6-8	-3% to -4%

- Starters out (snap-weighted, from injury reports) and the QB being out are real effects: those favorites won 67-72% instead of ~81%. Adding them as model features didn't help (65.1% -> 64.5% overall); as *eliminations* they add about 3 points.
- Favorites that "missed expectations" over their last 3 games (Gemini's rule) won 89% in validation, about the same as everyone else, so that filter removes good picks; it's not used for the NFL.
- Weather (wind, temperature, roof), rest and home-field advantage were already model inputs.
- The only way to 90%+ is agreeing with a heavy market favorite, priced around -950: break-even.

NCAA men's basketball (chosen on 2021-23, tested on 2024-26): favorites missing 15%+ of their regular minutes (players who sat out the last game) won 77.9% vs 83.3%. An 80% floor with the experts agreeing held **90.8% on 3,907 held-out picks (~1,300 a season)**; the roster filter is applied live as well.

WNBA: ~13 picks a season, too few to tune; dropping favorites in a slump went 89.6% vs 86.5% held-out (48 picks), not enough to adopt.

Live: `marv/stats/registry.py` now applies the NFL eliminations (market < 70%, 1.5+ starters out, wind 15+) and the NCAA men's 80% floor with the missing-minutes veto (`marv/data/roster.py`).

The owner's Pro Football Max formula: original vs fixed vs Marv (NFL 2016-2026)

`marv/stats/pfm.py`. Fixes: scoring drives fitted from real games (~6.2, not 2.4); a team's scoring rises with the opponent's EPA allowed (the original subtracted it); fitted home field (+1.8 pts); ties count half; point-in-time EPA blended with last season. Parameters fitted only on earlier seasons.

2,736 games	Winners	Brier	70%+ picks	78%+ picks	Avg projected total (miss)
Original formula	45.1%	0.264	none	none	16.8 (28.9 pts)
Fixed formula	63.3%	0.226	77.1% (358)	83.8% (68)	45.8 (10.8 pts)
Marv	64.5%	0.221	78.5% (395)	86.7% (83)	

Held-out 2022-26: fixed 63.2% vs Marv 65.4%; over/under vs closing totals 48.9% (no edge); the original veto (injuries, weather, tight spread) lowered 70%+ accuracy (74.7% vs 78.0%). The original's coin-flip-or-worse result comes from the defense sign. A 50/50 blend with Marv didn't beat Marv alone.

The owner's head-to-head total engine (pace, points per drive, red zone, Poisson)

On its own Cowboys–Bucs example it projects 78.7 points (line 47.5). Backtested on 2,736 NFL games 2016-2026 with real point-in-time inputs from nflverse play-by-play:

Model	Avg projected total	Avg miss	O/U vs closing total
Head-to-head engine as written	66.0	24.1	49.4%
Head-to-head engine fixed	47.4	11.7	49.2%
Pro Football Max fixed	45.8	10.8	48.3%
Marv	45.7	10.7	49.3% (51.1% when 3+ pts off the line, 570 games)
Closing total itself	45.2	10.4	-

Actual average 45.6; break-even at -110 is 52.4%. Problems in the original: drives = pace / 11.5 (~5.7 per team; real ~10-11) times a points-per-drive *rate* inflated by +1.8, then counted again as 3.2-point scoring events; subtracting the opponent's points allowed (a bad defense should add); and the red-zone multiplier double counts what points per drive already includes. No version beats the closing total: NFL totals stay a no-bet in Marv.

Normalized version (tuned 2016-21, tested 2022-26). Every input is relative to the previous season's league average and combined multiplicatively, `points = league_ppd x (offense/league)^a x (opponent_allowed/league)^b x (red_zone/league)^c x drives(pace)`, with a, b, c fitted. Fitted a = 0.55 (offense), b = 0.28 (defense); red zone and pace came out near zero (they add nothing beyond points per drive). Held-out: average projected total 43.6 (actual 45.0), miss 10.47 points (same as Marv; closing line 10.22), over/under vs closing totals 49.5% (50.2% when 4+ points off the line). Blending with the closing line was best at 100% market weight. Tethered to reality, but still no over/under edge.

Situational spots (prime-time overreaction, rest/travel, key numbers, divisional dogs)

Tested against closing spreads and real spread prices: 2006-15 (discovery) vs 2016-26 (confirmation).

Angle	2006-15	2016-26	Verdict
Fade last week's prime-time blowout winner	44.2% (156)	46.1% (154)	Backwards: backing them went 54.8% of 310 (p=0.19)
Back last week's prime-time blowout loser	48.8%	50.4%	No edge
Rested team, 4+ day rest edge	50.5%	51.6%	No edge
Home team vs West Coast visitor, 1:00 PM ET	52.1% (94)	44.0% (125)	Flipped; no edge
4+ day rest edge + 2 time zones of travel	48.3% (87)	54.3% (94)	Inconsistent
Divisional home dog +3.5 or more	48.6%	50.5%	No edge
Non-division home dog +3.5 or more	56.8% (229)	54.3% (258)	55.4% of 487 (p=0.09), but 46.8% in 2024-26
Dogs at +3 / +7 / +2.5 / +7.5	55.9 / 48.9 / 52.1 / 53.8%	51.9 / 49.0 / 52.1 / 53.8%	No stable key-number edge

Margins of exactly 3 (14.6%) and 7 (8.9%) are common, as the key-number idea says, but the closing prices around those numbers already reflect it. With ~15 angles tried, one or two $p < 0.10$ results are what luck alone produces. Nothing here qualifies as a bet trigger; the two best (prime-time blowout winners, non-division home dogs 3.5+) are worth tracking in paper mode only.

Advanced play-by-play features (EPA split, early-down success, explosives, pressure, EMA)

Built from nflverse play-by-play for every team-game 2009-2026, offense and defense: pass EPA per dropback, rush EPA per carry, early-down (1st/2nd) success rate, overall success rate, explosive rate (15+ yard passes, 10+ yard runs), pressure rate (QB hits + sacks per dropback, the closest nflverse has to pressure), EPA on pressured dropbacks. Recency-weighted with an EMA (half-life tuned on 2012-15: 6 games), only earlier games, matchup = my offense + their defense. Ridge model refit each season on the 6 before it.

Held-out 2016-26 (2,736 games)	Winners	Brier	70%+ picks	78%+ picks	ATS when 4+ pts off the spread
Advanced features	63.9%	0.2232	74.5% (593)	82.1% (196)	50.1% (587)
Marv	64.5%	0.2214	78.5% (395)	86.7% (83)	54.4% (463, p≈0.19)
50/50 blend	64.5%	0.2215	75.7% (469)	83.5% (115)	
Closing moneyline (no-vig)	66.5%	0.2108	79.0% (1,037)	82.5% (469)	

The stable efficiency stats don't beat Marv's full box-score set, and neither beats the market: the ~64% "side ceiling" is what public stats can reach; the closing line (66.5%) also prices injuries, weather and news. Not adopted.

Weighting each head-to-head stat

Instead of 1 point per stat won, a ridge regression learns each stat's weight from the 6 seasons before each test season ("learned weight per stat"), or also uses how big each edge is ("weights x size of edge").

Held-out	Equal weights	Learned weights	Weights x size of edge	Marv
NFL 2016-26 (2,641) winners	63.7%	63.9%	64.6%	64.7%
NFL 80%+ picks	83.2% (101)	91.7% (24)	91.8% (49)	91.5% (47)
College FB top 30, 2018-26 (1,432) winners	76.5%	76.9%	77.7%	77.7%
College FB 80%+ picks	88.4% (596)	87.2% (587)	88.9% (522)	92.2% (448)
ATS when 55%+ sure (NFL / CFB)	50.6% / 47.2%	51.5% / 48.4%	50.6% / 49.2%	

Weighting by the size of each edge adds about 1 point of winner accuracy over the 1-point-per-stat tally and brings the head-to-head method level with Marv, not past it (Marv's confident college picks stay better). In college football the heaviest weights went to points per drive, red-zone TD rate and defensive rush success / yards per play allowed; turnovers were weighted near zero (too noisy). Against the spread, no version has an edge.

marv/stats/h2h.py now supports `mode="weighted"` and `mode="magnitude"`.

Recalculating the head-to-head data points (weights, top stats, opponent strength, last-3 blends)

After learning each stat's weight, the head-to-head points were recalculated several ways and re-run through the walk-forward backtest (each season learned from the 6 before it, same games):

Held-out	Winners NFL / CFB	80%+ picks NFL / CFB	O/U every game NFL / CFB
Equal (1 point per stat)	63.7% / 76.5%	83.2% (101) / 88.4% (596)	
Weights x size of edge (current)	64.6% / 77.7%	91.8% (49) / 88.9% (522)	49.9% / 50.4%
Top 8 weighted stats only, refit	64.1% / 77.8%	82.3% (113) / 89.4% (530)	
Every stat adjusted for the opponents' strength	63.3% / 77.4%	84.6% (65) / 89.3% (561)	50.9% / 52.1%
75% season + 25% last 3	64.0% / 77.8%	88.7% (53) / 89.2% (536)	49.9% / 49.9%
75/25, opponent-adjusted	63.5% / 77.1%	89.2% (65) / 90.2% (572)	50.9% / 50.6%
Last 3 only, opponent-adjusted	60.9% / 73.5%	83.3% (18) / 88.7% (467)	50.5% / 50.6%
Marv	64.7% / 77.7%	91.5% (47) / 92.2% (448)	

- Winner differences are within the error bars (about ± 1.8 NFL, ± 2.2 college), except that last-3-only is clearly worse and opponent adjustment costs NFL 1.25 points (95% range -2.4 to -0.3).
- Opponent adjustment helps college calibration a little (Brier 0.163 -> 0.161, 80%+ picks up to 90%), since college schedules are uneven; in the NFL it adds noise.
- Pruning to the biggest weights doesn't help: ridge already shrinks the weak stats. NFL's top weights were turnover and kicking stats that change from season to season, a sign the NFL tally is mostly noise.
- Totals stay at 50-52% in every version, below the 52.4% needed at -110; projected totals miss by more than the closing line (NFL 10.8 vs 10.5, college 12.8-13.4 vs 12.5). Against the spread: 49-52%.

Verdict: keep weights x size of edge with the learned season + last-3 columns; no version passes Marv.
`h2h.walk_forward(..., adjust=1, blend=0.25)` and `h2h.breakdown()` (each stat's weighted points for a game) are available for paper tracking.

Recency weighting: last 25% of the season, exponential decay

Gemini suggested weighting the last 25% of the season (or a compressed rolling window) into the Monte Carlo, especially for totals. Tested on the same held-out games, magnitude weights, raw and opponent-adjusted (scratch script; "last 25%" = the most recent quarter of the games a team has played this season, at least 2):

Held-out	Winners NFL / CFB	80%+ picks NFL / CFB	Spread 55%+ NFL / CFB	O/U NFL / CFB
Season + last 3 (current)	64.6% / 77.7%	91.8% (49) / 88.9% (522)	50.6% / 49.2%	49.9% / 50.4%
Season + last 25%	64.2% / 77.8%	87.2% (47) / 88.9% (515)	51.1% / 50.2%	49.6% / 51.0%
75% season + 25% last quarter	63.8% / 77.2%	90.7% (43) / 88.4% (510)	50.3% / 48.7%	49.7% / 50.2%
Exponential, half-life 3 games	61.7% / 74.7%	86.4% (22) / 88.7% (433)	49.4% / 47.2%	49.3% / 50.9%
Season + last 25%, opponent-adjusted	63.3% / 77.7%	84.1% (63) / 90.9% (540)	50.3% / 51.2%	49.8% / 50.7%
75/25 last quarter, opponent-adjusted	63.7% / 77.7%	89.1% (64) / 90.5% (535)	50.6% / 50.2%	50.7% / 50.8%
Exponential, opponent-adjusted	61.7% / 75.5%	91.7% (36) / 90.0% (478)	49.6% / 47.8%	49.9% / 50.4%

No recency version beats the current one on winners, and none gets the spread or totals to 52.4%. Heavy recency (exponential decay) is clearly worse: a few games are too small a sample. Projected totals miss by more than the closing line in every version. Not adopted for live O/U; the one mild positive (college 80%+ picks at ~91% with opponent adjustment) is still below Marv's 92.2%.

Team over/under trends and the trend Monte Carlo vs closing totals

Does a team's recent over/under record, or the head-to-head Monte Carlo total built from recent-form stats, predict the next game's result against the closing total? NFL 2010-26 (4,382 games with a closing total), college FB 2017-26 (6,727 FBS games); Monte Carlo totals are walk-forward predictions.

Rule (win rate vs the closing total, -110)	College FB	NFL
Both teams over in 2+ of their last 3 -> bet OVER	45.1% (994)	49.7% (849)
Same games -> bet UNDER (fade the trend)	54.9% (994, p=0.002)	50.3%
Same, games with a top-30 team	56.6% (325)	
Both teams' last 3 beat the total by 14+ combined -> UNDER	54.6% (683)	51.2% (422)
Both teams' season over rate 60%+ -> OVER	42.7% (314)	45.8% (297)

Rule (win rate vs the closing total, -110)	College FB	NFL
Monte Carlo total (season + last 3), 55%+ sure	51.5% (3,369)	48.8% (1,450)
Monte Carlo total (last 3, opponent-adjusted), 55%+ sure	50.4% (3,780)	50.4% (1,663)
Monte Carlo agrees with the team trend	44.7% (1,039)	49.5% (610)
Monte Carlo disagrees with the trend -> follow Monte Carlo	52.6% (821)	52.3% (501)

- Following a team's over/under trend loses. In college football the opposite works: the more overs both teams had in their last 3 combined, the more often the next game went under (0/6 overs: 40% under, 3/6: 51%, 5/6: 57%, 6/6: 63%). The closing total seems to over-adjust to recent high-scoring games. The fade beat 52.4% in 8 of 9 full seasons (2024 was 48.5%), and same-week random games went under only 50.9% (p=0.008).
- The NFL shows no such pattern; its totals are sharper.
- The trend Monte Carlo has no totals edge by itself, and when it agrees with a team's streak it loses, because it is chasing the same recent scoring the line already priced.
- Caveat: ~24 rules were tested on two sports, so the best one is partly selected by luck; prices are assumed -110 (college juice varies). Paper-track "fade both-teams-over-trending college totals" before staking anything.

Continuation or reversal: market over-adjustment, scoring luck and pace

Gemini's idea: an over/under trend continues when it's backed by volume and the market under-reacted, and reverses when the market moved the total more than the underlying efficiency justifies. Tested per game, pre-game only: market move = this closing total minus the average total of each team's earlier games (last season worth 4 games + this season's games before the last 3); yardage move = the change in the teams' game yardage over the last 3 x league points per yard (0.069 college, 0.063 NFL); over-adjustment = market move minus yardage move. Scoring luck = points beyond what the yardage usually produces. College 4,024 games, NFL 3,767 (needs 3+ earlier games that season).

Rule (win rate vs the closing total)	College FB	NFL
Over trend (both teams 2+ overs in last 3) -> UNDER	54.4% (924)	51.2% (916)
Over trend + market over-adjusted (2+ pts beyond yardage) -> UNDER	56.5% (310)	54.3% (234; 50.0% early, 58.6% late)
Over trend + market matched -> OVER (continuation)	46.6% (614)	49.9% (682)
Over trend from scoring luck -> UNDER	55.0% (462)	53.5% (458)
Over trend from pace (plays up 5+) -> OVER	42.6% (216)	42.4% (132)
Under trend (both 0-1 overs in last 3) -> OVER	54.3% (1,120)	48.8% (1,018)
Under trend + market matched -> UNDER (continuation)	45.3% (691)	50.5% (792)
Any game: total moved 5+ pts more than yardage justifies -> UNDER	52.2% (737)	55.8% (400; 55.4% / 56.1% by half)

- “Continuation” never paid: in college football both over and under trends reverse whether or not the market over-adjusted. Over-adjustment and scoring luck only make the reversal stronger.
- Pace doesn’t make a trend sticky either: over trends driven by more plays went over only 42-43% next.
- NFL: the trend itself has no edge, but a total that rose 5+ points more than recent yardage justifies went under 55.8% (p=0.024), steady in both halves of 2009-26.
- About 30 rules were tested here; the leads worth paper-tracking are college “fade both trends” and NFL “total inflated 5+ beyond yardage -> UNDER”. Neither is proven.

Line movement, Bovado vs sharp books, and closing-line value (college football)

Data: opening and closing lines from ~25 books for every FBS game (sportsdataverse cfbfastR-data, `marv/data/cfb_lines.py`); opening lines for 8,971 games with results 2014-25 (none in 2020), Pinnacle / BetCRIS closing numbers 2014-19, Bovado lines and prices throughout. Ticket and money percentages (betting splits) have no free history anywhere, so public-vs-sharp splits can’t be backtested; line movement is the measurable trace of that money.

Test	Win rate (n)	Notes
Follow a 2+ pt spread move, bet at the close	48.7% (3,136)	fading: 51.3%
Favorite's number shrank 1.5+ (“money on the dog”) -> dog at the close	49.6% (2,038)	reverse-line-move proxy: no edge
Total fell 1.5+ -> UNDER at the close	52.0% (2,811)	53.7% to 2019, 50.1% since 2021
Bet the open on the side the spread later moved toward	55.8% (5,639)	only if you can predict the move
Bet the open on the side the total later moved toward	58.3% (4,917)	same
Spread: Bovado 0.5+ pts better than Pinnacle/BetCRIS -> bet it at Bovado	53.2% (2,429)	ROI +0.6% after Bovado's real juice
Total: Bovado 0.5+ pts off Pinnacle/BetCRIS -> better side at Bovado	54.4% (2,340)	ROI +4.6% (2014-19)
Same spread side graded at the sharp number (control)	48.7% (842)	the edge is the number difference
Spread moved toward the head-to-head model's side (model 3+ off the open)	56.0% (3,161)	“closing-line value”
...but the model's side against the spread at the open / close	50.2% / 49.4%	the CLV doesn't turn into wins
Over trend (both teams 2+ overs in last 3) -> UNDER	53.5% (1,829)	ROI +2.1%
Over trend + total rose 1.5+ from the open -> UNDER	56.7% (298)	ROI +8.3%; 58.6% to 2019, 54.3% since
Under trend -> OVER	51.2% (2,050)	weaker with consensus closes than in the earlier test

- Moves are fully priced by the close: following or fading them at the closing number does nothing. The value of a move is only available before it happens (55.8% ATS at the open).
- The model's picks show closing-line value (lines move toward it 56% of the time) yet don't win at the open. The markets move toward public stats that the model shares; the moves that matter (injuries, news) aren't in its inputs. CLV alone can overstate a stats model.

- The one usable consensus signal is line shopping against the sharp books: when Bovado's total is half a point or more off Pinnacle/BetCRIS, the better side at Bovado went 54.4% (+4.6% ROI). Spreads were break-even after Bovado's juice. Live Pinnacle prices are in The Odds API's "eu" region.
- The over-trend fade gets stronger when the total has also been bid up during the week (56.7%), which fits the market over-adjusting to recent shootouts. ~40 rules tested in this section: paper-track.
- NFL: no free opening-line history; on the VM, The Odds API historical endpoint can supply open and close (credits).

Backup-QB flag (NFL)

Every NFL game where either team's latest starting QB has fewer than 3 starts (this and last season, any team) or its starting QB is ruled out is now skipped for every bet. Backtest 2016-26 (nflverse starters):

	Games	Winners	78%+ picks	Marv 4+ pts off the spread, ATS
Normal starters	2,308	64.2%	87.1% (62)	54.3% (302)
Backup QB in the game	507	65.9%	85.7% (21)	54.4% (169)

Marv disagrees with the spread more in backup-QB games (3.4 vs 2.2 points on average), but its results in those games were no worse, so the flag isn't a proven improvement: it's a safety rule against betting into injury news the stats lag behind (e.g. Tampa Bay's Jalon Daniels, Oct 8 2026: market Dallas -8.5 to -10, Marv Dallas by 3).

NFL simulator calibration (key numbers and spread of outcomes)

Every 2016-26 game simulated at its closing spread and total, compared with real finishes:

	Old sim	Calibrated sim	Real games
Games decided by exactly 3	8.0%	~13-14%	14.7%
Decided by exactly 7	6.6%	~8.5%	8.5%
Spread of totals around the closing line (SD)	14.2	13.3	13.2
Spread of margins around the closing spread (SD)	13.8	12.7	12.7

Changes ([marv/sims/football.py](#), NFL only): real extra-point outcomes (7/6/8), a last-possession phase that plays the score (field goal when tied or down 1-3, touchdown when down 4-8), NFL overtime (57% end by 3, 6% tie), less game-to-game "form" swing (shape 30 -> 120), and a calibrated transfer of near-miss finishes onto 3 and 7.

Backtest with Marv's own projections (2,815 games, 2016-26):

	Old	Calibrated
Moneyline Brier	0.2221	0.2219
Push probability on spreads of 3 (real 10.2%)	4.5%	7.1%
Push probability on spreads of 7 (real 6.4%)	3.8%	4.8%
Against the spread, log loss (coin flip = 0.693)	0.7048	0.7086
Over/under, log loss	0.7033	0.7044
ATS when 55%+ sure	51.2% (1,437)	51.8% (1,628)

The simulator now reproduces real NFL scores and prices pushes far better, which matters for spreads on 3 and 7, live prices and quarter bets. It does not make Marv's spread or total picks better: with either simulator they are worse than a coin flip in log loss, because Marv's projections disagree with the closing line mostly where the line is right. NFL spreads and totals stay off as picks.

NFL player props vs real sportsbook lines (2025)

props-backtest --seasons 2025 --real --max-credits 6000: real Bovado/DraftKings/FanDuel lines one hour before kickoff, 4,128 priced props (credit cap reached partway through the season).

Market	Edge 0%+	Edge 5%+	Edge 10%+
All markets	49.1% of 3,040, ROI -5.5%	48.7% of 1,640, ROI -5.1%	48.8% of 857, ROI -4.6%
Passing yards	51.7% (201)	46.3% (95)	48.9% (47)
Receiving yards	50.3% (1,132)	50.3% (581)	51.9% (289), ROI -0.9%
Receptions	46.2% (1,074)	45.5% (525)	41.5% (234), ROI -11.7%
Rushing yards	51.2% (633)	50.8% (439)	51.6% (287), ROI -2.5%

Always betting the under went 52.5%, better than the model. The 55-63% earlier came from lines built from averages; real books set props near the median and price in usage and matchups. Bigger model edges did not win more often. No prop edge: prop cards are off by default (PROPS=on to restore them).

Market-anchored confidence and where the pick engine can (and can't) improve

Walk-forward, expanding window, held-out seasons (NFL 2018-25 n=2,219; college top-30 2018-25 n=2,336; logistic fits on earlier seasons only). Script: scratchpad stack.py, stack2.py, stack3.py.

Question	Result
Does the model add information beyond the closing spread? (stacked logistic)	No. NFL model coefficient 0.00-0.01 vs market 0.14; CFB 0.00 vs 0.12. Brier: market 0.2108, stack 0.2109, model alone 0.2204
Does it add to the total? (actual total – line regressed on model total – line)	No. NFL weight –0.20, CFB –0.02; top-20% edge picks hit 50.2% / 50.4%. NBA lines in the file are unreliable so NBA was not judged
Moneyline accuracy at equal volume (NFL, top 400 by confidence)	model alone 76.5% (ROI –6.6%), market 84.3% (–0.4%), 20% model + 80% market 84.8% (+0.2%)
Same, ≥85% confidence	NFL 88.6-90.5%, CFB 93.4% (900 games) vs 88.6% / 91.5% for the model alone

What changed: stats-mode moneyline confidence is now $0.2 \times \text{model} + 0.8 \times \text{no-vig market}$ (StatsRules.ml_model_weight, NFL and college football; floors 0.85). Accuracy at a given volume goes up because the engine stops treating model noise as confidence on games the market already prices at 70-90%.

What did not and cannot change: this is not new skill. It's the market's information, so ROI stays break-even (–2% to +0.2%), and over/unders have no edge from the model (50%). Gains that can still be tested: college basketball and WNBA (need the VM's line history to run the same stack), injury/lineup news the market hasn't priced, and the paper-tracked trend tags (college O/U fades, Bovado-vs-Pinnacle totals).